Dr. Amir Goldental

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Summary

A passionate problem solver with a decade of experience in research and development in cutting edge interdisciplinary fields. I am dedicated to consumer-facing solutions that make an impact. My experience ranges from theoretical discoveries in the realm of mathematics to product development and deployment. **I am a leader that can consolidate a vision and realize it.**

Professional Experience

2016 - Ongoing: **Research Scientist and Developer**, Aerospace systems, Ministry of Defense.

- **Understanding the needs of the end users and developing dedicated solutions.**
- **Research and development** of algorithms for physical modeling and signal processing. Applying methods from the worlds of data science, optimization and deep learning.
- **Software development** and architecture with distributed computing.
- Experience as **team lead.**
- Full control over product life cycle, **from inception to deployment** and support.
- Military service.

2012 - Ongoing: **Interdisciplinary research** in the Physics department and the brain research center of Bar-Ilan University**.**

- **19 peer-reviewed publications** in the fields of Physics, Mathematics, Smart Transportation and experimental neuroscience. (See attached).

2014 - 2019: **Academic teaching** at the Physics department of Bar-Ilan University.

- "Information theory and machine learning".
- **"**Advanced statistical mechanics" for Physics M.Sc. students.

Education

2018 - Ongoing: **Postdoctoral Fellowship** at the Bar-Ilan University.

- Research in the field of complex systems, optimization and autonomous vehicles.
- Development of multi-agent traffic flow simulator for research and prediction.

2014 - 2018: **Direct Ph.D. track in theoretical physics**, Bar-Ilan University. **With honors.**

- Mathematical research, simulations of complex systems, modeling, neural networks, embedded software development, experimental neuroscience.

2011 - 2014: **B.Sc. in Physics**, Bar-Ilan University. **With honors.**

- Average grade: **96, During high school.**

Technical Skills & Expertise

Languages and frameworks: Python (ML, Math and Web libraries), Matlab, C++.

Technical skills: Optimization, Machine learning, Software development, Image and signal processing, Statistics, Complex systems.

Disciplines: Data science, Aerospace, Physics, Neuroscience, Smart transportation.

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Awards and Honors

- 2021: Ministry of Defense excellence **award**.
- 2018: Wolf Foundation **scholarship** for excellent PhD student.
- 2018: One of the **30 most promising** people under the age of 30 in Israel, **Forbes**.
- 2016: One of the **40 most promising** people under the age of 40, **TheMarker**.
- 2016: Rector's **award** for outstanding PhD students.
- 2016: Invited to The Lindau **Nobel Laureate Meetings**.
- 2015: Best presentation **award**, Japan's HOPE meeting with Nobel laureates.
- 2014 - 2017: President's **scholarships** for outstanding doctoral students.
- 2013: Ashdod's municipal **scholarship** for outstanding university students.
- 2013: Second place in the Bar-Ilan's mathematics **competition**.
- 2012 - 2014: "Outstanding in science" **scholarship** of Bar Ilan's Institute for Nanotechnology and Advanced Materials (BINA).

Presentations in Conferences and Meetings

- Ein Gedi conference of The Leslie and Susan Gonda Multidisciplinary Brain Research Center. **Oral presentation** (2014)**:** "An experimental evidence-based computational paradigm for new logic-gates in neuronal activity".
- 60th annual meeting of the Israel Physical Society. **Oral presentation** (2014)**:** "A computational paradigm for dynamic logic gates in neuronal activity".
- 7th HOPE Meeting with Nobel Laureates in Japan. **Poster** (2015)**:** "Low firing rates and microsecond precision in the brain".
- Kyoto university, physics department. **Colloquium** (2015): "Stability of activity in neural networks".
- Meeting in Gottingen University (2016). **Oral presentation:** "Optimizing treatment for brain disorders".
- Association of Engineers, Architects and Graduates in Technological Sciences in Israel. **Webinar** (2020): "Improving traffic flow with the help of cooperating autonomous vehicles".

Publication List

1. An experimental evidence-based computational paradigm for new logic-gates in neuronal activity.
Vardi R, Guberman S, **Goldental A**, and Kanter I.
Europhysics Letters 103, 66001 (2013).

2. Sudden synchrony leaps accompanied by frequency multiplications in neuronal activity.
Vardi R*, Guberman S*, **Goldental A***, Kalmanovich A, Marmari H and Kanter I.
Frontiers in Neural Circuits 7, 176 (2013).

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Publication List (Continued)

3. A computational paradigm for dynamic logic-gates in neuronal activity.
Goldental A*, Guberman S*, Vardi R* and Kanter I.
Frontiers in Computational Neuroscience 8, 52 (2014).

4. Neuronal impedance mechanism implementing cooperative networks with low firing rates and μs precision.
Vardi R*, **Goldental A***, Marmari H, Brama H, Stern E, Sardi S, and Kanter I.
Frontiers in Neural Circuits 9, 29 (2015).

5. Broadband macroscopic cortical oscillations emerge from intrinsic neuronal response failures.
Goldental A*, Vardi R*, Sardi S, Sabo P and Kanter I.
Frontiers Neural Circuits 9, 65 (2015).

6. Hours of high-frequency stimulations reveal intracellular neuronal trends in vivo.
Brama H*, **Goldental A***, Vardi R, Stern E, and Kanter I.
Europhysics Letters 116, 46002 (2016).

7. Mimicking collective firing patterns of hundreds of connected neurons using a single-neuron experiment.
Goldental A*, Sabo P*, Sardi S, Vardi R* and Kanter I.
Frontiers in Neuroscience 9, 508 (2016).

8. Vitality of neural networks under reoccurring catastrophic failures.
Sardi S*, **Goldental A***, Amir H, Vardi R and Kanter I.
Scientific Reports 6, 31674. (2016).

9. Simultaneous multi-patch-clamp and extracellular-array recordings: Single neuron reflects network activity.
Vardi R*, **Goldental A***, Sardi S, Sheinin A and Kanter I.
Scientific Reports 6, 36228 (2016).

10. Oscillations in networks of networks stem from adaptive nodes with memory.
Goldental A*, Uzan H*, Sardi S and Kanter I.
Scientific Reports 7, 2700 (2017).

11. Fast reversible learning based on neurons functioning as anisotropic multiplex hubs.
Vardi R*, **Goldental A***, Sheinin A, Sardi S and Kanter I.
Europhysics Letters 118, 46002 (2017).

12. New types of experiments reveal that a neuron functions as multiple independent threshold units.
Sardi S, Vardi R, Sheinin A, **Goldental A** and Kanter I.
Scientific Reports 7, 18036 (2017).

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Publication List (Continued)

13. Adaptive nodes enrich nonlinear cooperative learning beyond traditional adaptation by links.
Sardi S*, Vardi R*, **Goldental A***, Sheinin A, Uzan H and Kanter I.
Scientific reports 8, 5100 (2018).

14. Less might be more: Conduction failure as a factor possibly limiting the efficacy of higher frequencies in rTMS protocols.
Halawa I, **Goldental A**, Shirota Y, Kanter I, Paulus W.
Frontiers in neuroscience 12, 358 (2018).

15. Stationary log-normal distribution of weights stems from spontaneous ordering in adaptive node networks.
Uzan H, Sardi S, **Goldental A**, Vardi R, Kanter I.
Scientific reports 8, 13091 (2018).

16. Biological learning curves outperform existing ones in artificial intelligence algorithms.
Uzan H, Sardi S, **Goldental A**, Vardi R, Kanter I
Scientific reports 9 (1), 1-11 (2019).

17. Brain experiments imply adaptation mechanisms which outperform common AI learning algorithms.
Sardi S, Vardi R, Meir Y, Tugendhaft Y, Hodassman S, **Goldental A**, Kanter I.
Scientific Reports 10 (1), 1-10 (2020).

18. A minority of self-organizing autonomous vehicles significantly increase freeway traffic flow.
Goldental A, Kanter I.
Journal of Physics A: Mathematical and Theoretical 53 (41), 414001 (2020).

19. Power-law scaling to assist with key challenges in artificial intelligence.
Meir Y, Sardi S, Hodassman S, Kisos K, Ben-Noam I, **Goldental A**, Kanter I.
Scientific reports 10 (1), 1-7 (2020).

** These authors contributed equally to this work.*